SHIMAA ELMORSHEDY

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PROFESSIONAL OBJECTIVE

AI and Machine Learning professional with hands-on experience in Python, Machine Learning, Deep Learning, and Natural Language Processing. Passionate about applying AI solutions and teaching concepts through practical projects. Strong technical, analytical, and problem-solving skills with experience in full ML/DL project lifecycles.

EDUCATION

Sadat City University — Egypt Feb 2023 – Jan 2027

Bachelor of Science in Computer Science and Artificial Intelligence

PRACTICAL TRAINING & EXPERIENCE

National Telecommunication Institute (NTI) *NLP Trainee*

- Studied practical Natural Language Processing (NLP) concepts including text preprocessing, tokenization, text classification, TF-IDF, and word embeddings.
- Developed sequence-based NLP models and implemented text classification pipelines.

Information Technology Institute (ITI) *Backend .NET Trainee*

- Received intensive practical training in Backend Development using C# and .NET frameworks.
- Applied relational database management, Object-Oriented Programming (OOP), and backend architectural concepts.

EverGrow *IT Department Trainee*

- Gained practical exposure to enterprise IT operations, infrastructure troubleshooting, and working within a professional technical environment.

KEY PROJECTS

Bone Fracture Classification using Deep Learning

- Developed a CNN-based medical image classification system for bone fracture detection from X-ray images using CLAHE preprocessing and data augmentation.
- Achieved 94% classification accuracy, outperforming baseline models by 8%.
- Deployed an interactive Streamlit web application integrating the complete preprocessing and inference pipeline for real-time predictions.

Email Spam Detection System

- Developed an NLP-based email classification model utilizing TF-IDF vectorization trained on a dataset of 5,000+ emails. Achieved 96% precision and 94% recall in spam detection; deployed as a real-time Streamlit web application.
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KEY PROJECTS (CONTINUED)

Job Description Search Engine

- Implemented an Information Retrieval (IR) system using TF-IDF, inverted indexing, and cosine similarity to index and rank 1,000+ job descriptions.
- Built a searchable web interface returning top-matched results with similarity scores, achieving sub-second query response time.

Student Enrollment Prediction

- Built and evaluated 4 predictive ML models (Logistic Regression, Decision Tree, Random Forest, SVM) to forecast student enrollment decisions across 500+ records.
- Achieved 87% accuracy with Random Forest, outperforming baseline Logistic Regression by 12%.

TECHNICAL SKILLS

Programming & Tools:	Python (OOP), C#, .NET, SQL, Git, OpenCV, Streamlit
Machine Learning & AI:	Scikit-Learn, XGBoost, Supervised Learning, Unsupervised Learning, Model Evaluation
Deep Learning:	TensorFlow, Keras, Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), Transfer Learning, Optimization Techniques
NLP & IR:	TF-IDF, Inverted Indexing, Cosine Similarity, Word2Vec, GloVe, FastText, Text Classification, Sentiment Analysis, RNN, LSTM, GRU, Transformers
Data Science & Analytics:	Pandas, NumPy, Matplotlib, Seaborn, Data Cleaning, Preprocessing, Exploratory Data Analysis (EDA)

CERTIFICATIONS

- **Machine Learning Specialization** — Coursera (DeepLearning.AI)
- **Databases and SQL for Data Science with Python** — Coursera (IBM)
- **HCIA-AI V4.0 Certification Course** — Huawei ICT Academy
- **Machine and Deep Learning Training** — NTI
- **Data Analyst in Python** — DataCamp
- **Natural Language Processing (NLP) in Python** — DataCamp
- **AI For Everyone (AI4E)** — ITI (Mahara-Tech)
- **Build with AI: Masr Edition** — Google x ITI

LANGUAGES

- **Arabic**: Native
- **English:** Fluent